

Interface Control Document (ICD)

For Electronic Safe and Arm Device (ESAD)

Board-Level Prototype

For the Integration onto UDS's Avenger 3 UAV

to Excalibur International

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ABBREVIATIONS

BIT	: Built-In Test
EFI	: Exploding Foil Initiator
ESAD	: Electronic Safe and Arm Device
HEX	: Hexadecimal
IO	: Input Output
LEEFI	: Low Energy Exploding Foil Initiator
LSB	: Least-Significant Bit
MSB	: Most-Significant Bit
NA	: Not Applicable
OC	: Onboard Computer
UAV	: Unmanned Air Vehicle

1 GENERAL

1.1 SCOPE

This document defines and controls the interfaces between the Electronic Safe and Arm Device (ESAD) and the UAV platform for integration of a HE warhead. It establishes the technical and operational interface parameters, describing both electrical and mechanical (and where applicable, software) interactions. The goal is to ensure safe, secure, and reliable functioning of the warhead's ESAD, including the incorporation of proximity and impact switch functionalities, throughout the system's lifecycle.

1.2 SYSTEM DEFINITION

In general, an electronic safe and arm device is a high-voltage, all-electronic safety device used to control in-line the initiation sequence of a pyrotechnic train on warhead modules. Essentially, it is a safety device used to prevent the accidental arming or firing of a pyrotechnic event by eliminating system sensitivities to environmental stimuli, such as electromagnetic fields, electrostatic discharge, and stray voltages.

Here the Artron's offered generic Electronic Safe and Arm Device (ESAD) solution provides safety critical energetic systems initiation on command, when commanded for warhead operations and activates the warhead upon impact. Designed for applications where a reliable arming and firing sequence with precise timing events is vital. The ESAD described here is compatible with the EFI / LEEFI type detonator. Depending on the final chosen detonator manufacturer, mechanical interface with the detonator part is subject to revision.

ESAD has Input Power interface and Communication interfaces with Onboard Computer / Flight Control Computer. It has also interfaces with Umbilical connection, warhead LEEFI detonator and proximity sensor sub-units.

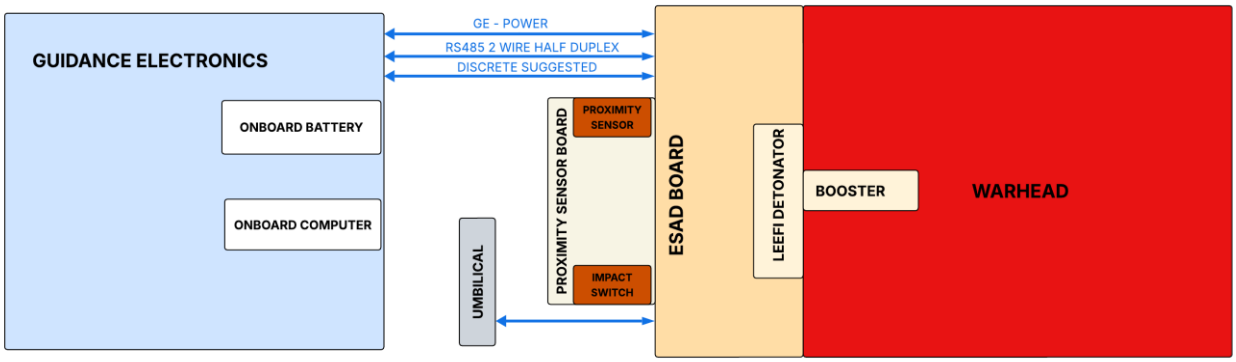


Figure 1: ESAD Board System Interfaces

The key features of the ESAD can be listed as follows:

- Design compatible with STANAG 4187 – Fuze Safety Design Requirements
- Design compatible with MIL-STD-461 for EMI/EMC.
- Customizable safety logic, safe separation timers, break wires, acceleration set points, and I/O configurations
- Alternative mechanical packaging options in coordination with the upper System's designers.

- Proximity sensor board should be placed in front face of the upper system, separated from the ESAD unit.
- High dynamic and thermal environmental qualification heritage
- Internal high voltage generation for Low Energy Exploding Foil Initiators
- Static and dynamic switches for safety checks
- Power on self-test and arm status outputs

1.3 ESAD OPERATIONAL FLOW SEQUENCE

Main Operational Flowchart for ESAD is given in below Figure. Draft version of the detailed ESAD Operational Flow Sequence is given show in figure.

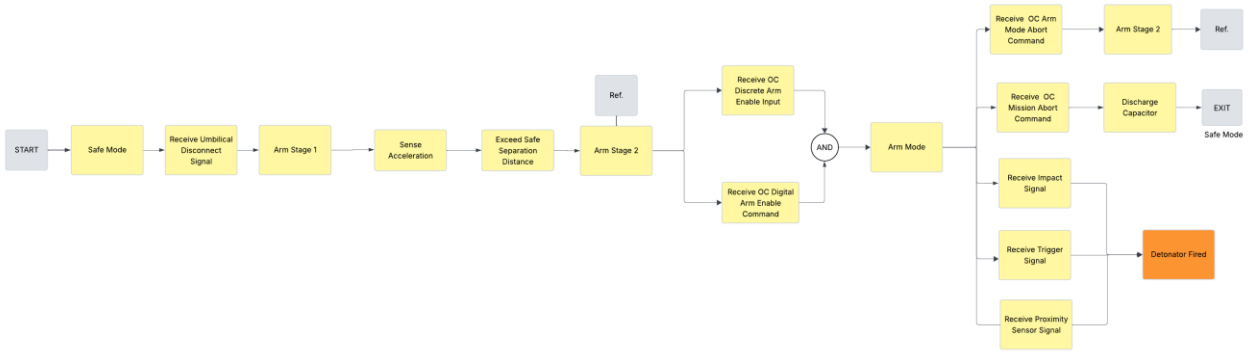
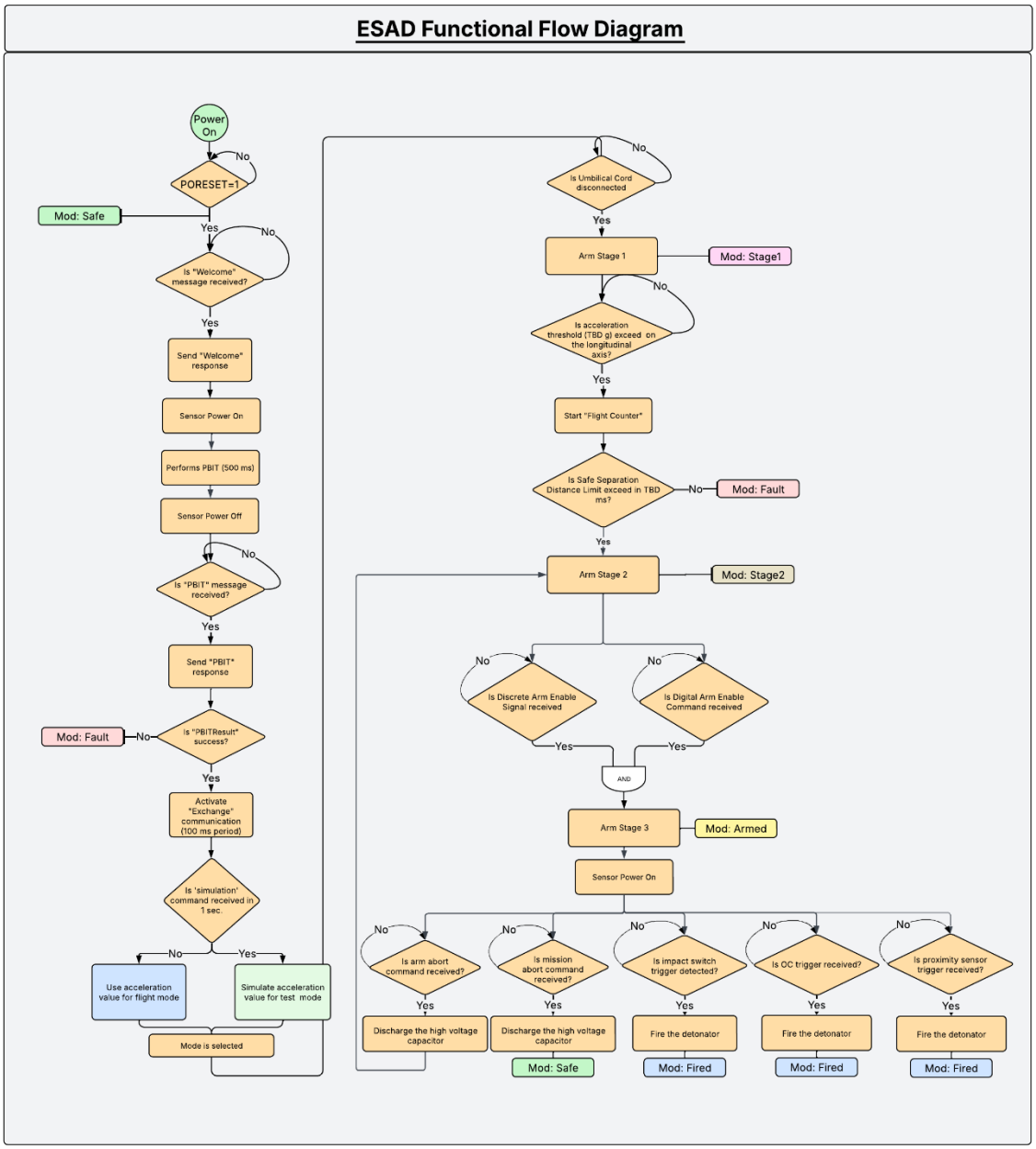


Figure 2: ESAD Operational Block Diagram



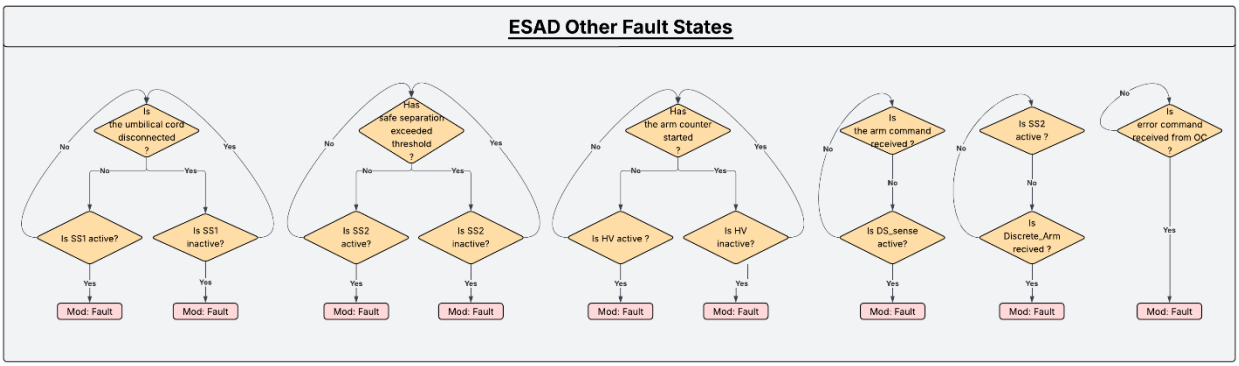


Figure 3: ESAD Software Flowchart

1.4 ESAD INTERFACES

1.4.1 Electrical Interface

This section describes the electrical interfaces of ESAD and Sensor Boards. There are discrete signals between ESAD and Onboard Computer (communication and discrete input), Umbilical connection (if available) and LEEFI detonator. Also there is an interface for ESAD Board and Sensor Board

ESAD Board has three dedicated interface connectors (J100 for System, J102 for Sensor Board, J401 for Detonator) and one reserved connector (J101). Pinout tables are as shown in the section below. NA pins are not used for the generic configuration, but can be used in accordance with the system design when necessary. Depending on the system design of the main platform that the ESAD will be integrated onto, it is always possible to revise the interfaces and operational sequence of the ESAD.

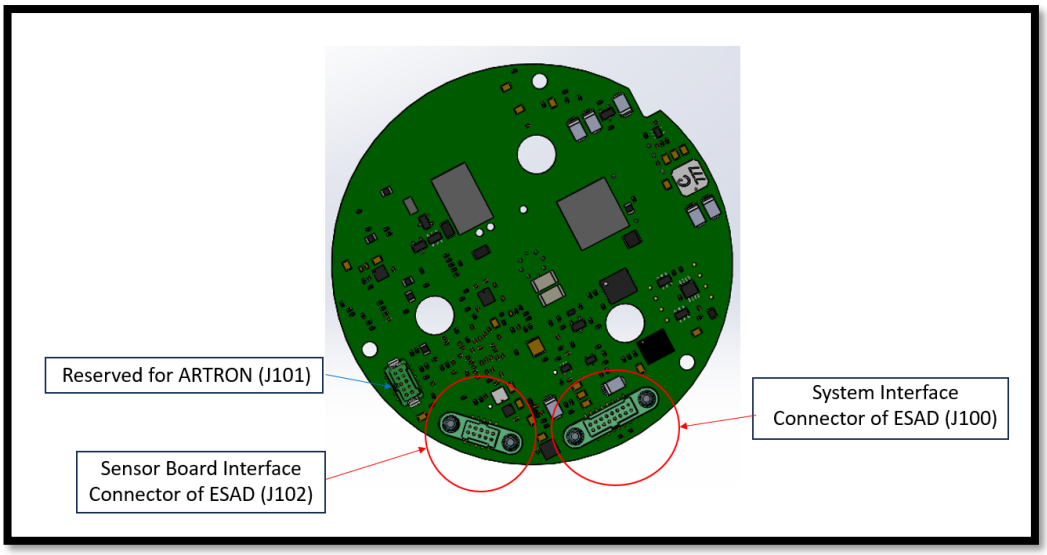


Figure 4: ESAD Board Connectors

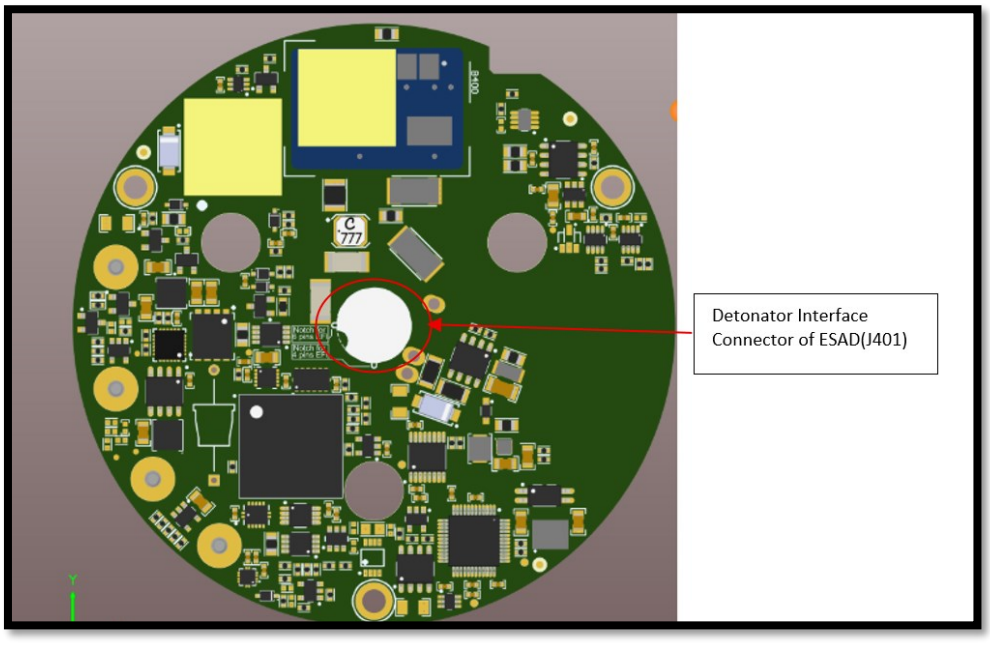


Figure 5: ESAD Board Connectors

Sensor Board has one USER dedicated interface connector (J1 for ESAD Board) and one ARTRON reserved connector (J2). Pinout tables are as shown in the section below.

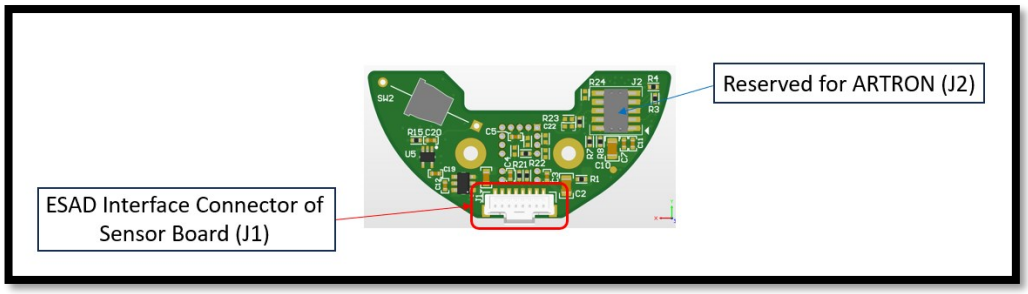


Figure 6: Sensor Board Connectors

1.4.1.1 J100 Connector on ESAD Board

J100 connector on ESAD Board is used for System interface connector. The mating connector model number is G125-2241696F1 and crimps model number is G125-0020005 from Harwin.

Table 1 J100 Connector Suggested Pinout

Pin No	Signal Name	Description
1	VDC_IN	ESAD Power Input
2	VDC_IN	ESAD Power Input
3	GND	Common Ground
4	GND	Common Ground
5	UMB	Umbilical Continuity Line.
6	UMB_RTN	Umbilical Continuity Return Line.
7	NA	-
8	NA	-
9	RS485_AB_P	RS-485 Communication Interface (Differential Positive Line)
10	RS485_AB_N	RS-485 Communication Interface (Differential Negative Line)
11	EXT_RS485_P	RS-485 Communication Interface (Differential Positive Line) (Spare)
12	EXT_RS485_N	RS-485 Communication Interface (Differential Negative Line) (Spare)
13	DSC_IN1	Arm Enable Discrete Input Signal.
14	DSC_IN2	Discrete Input 2
15	DSC_OUT1	Discrete Output 1
16	DSC_OUT2	Discrete Output 2

1.4.1.2 J102 Connector on ESAD Board

J102 connector on ESAD Board is used for Sensor Board interface connector. The mating connector model number is G125-2241096F1 and crimps model number is G125-0020005 from Harwin.

Table 2 J012 Connector Suggested Pinout

Pin No	Signal Name	Description
1	DIST_SPARE	Spare Line
2	DIST_COMM1	Communication Line (UART RX) (Receiver for ESAD)
3	DIST_COMM2	Communication Line (UART TX) (Transmitter for ESAD)
4	IRQ_L	Sensor Trigger Signal
5	NA	-
6	VOUT	Sensor Board Power
7	VOUT	Sensor Board Power
8	GND	Common Ground
9	GND	Common Ground
10	NA	-

1.4.1.3 J401 Connector on ESAD Board

J401 connector on ESAD Board is used for detonator custom interface.

Table 3 J401 Connector Pinout

Pin No	Signal Name	Description
1,2,3	HV_OUT	High Voltage
5,6,7	HV_GND	High Voltage Ground
4,8	NA	- (Pins removed)

1.4.2 Necessary Mating Connector Pinouts

ESAD Board's J100 mating connector (Harwin / G125-2241696F1) pinout is given below.

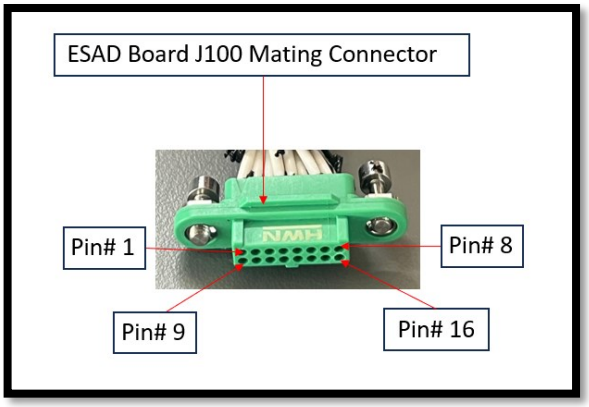


Figure 7: ESAD Board J100 16-Pin Mating Connector Pinout

ESAD Board's J102 mating connector (Harwin / G125-2241096F1) pinout is given below.

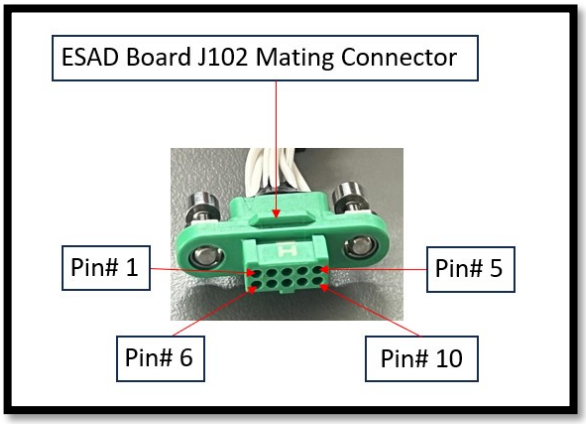


Figure 8: ESAD Board J102 10-Pin Mating Connector Pinout

Sensor Board's J1 mating connector (Molex / 501330-0800) pinout is given below.

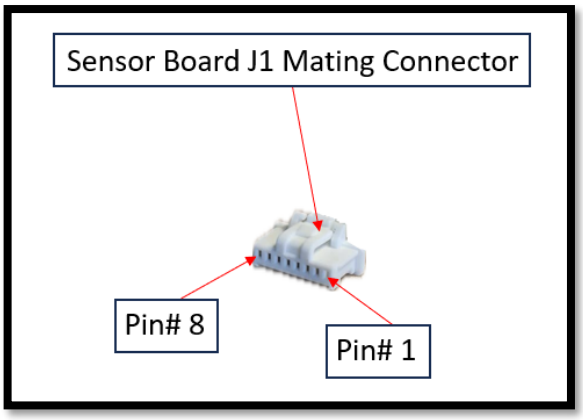


Figure 9: Sensor Board J1 8-Pin Mating Connector Pinout

1.4.3 Cabling Between ESAD Board J102 Connector and Sensor Board J1 Connector

The cabling between Sensor Board J1 Connector and ESAD Board J102 connector should be made due to the table below. Pinout information of mating connectors are given with **Figure 9** and **Figure 8**.

Sensor Board J1 Mating Connector (Molex / 501330-0800)		Cable AWG	ESAD Board J102 Mating Connector (Harwin / G125-2241096F1)	
Pin #	Signal Name		Pin #	Signal Name
1	DIST_SPARE	28AWG	1	DIST_SPARE
2	IR_ESAD_L	28AWG	4	IRQ_L
3	DIST_COMM2	28AWG	2	DIST_COMM1
4	DIST_COMM1	28AWG	3	DIST_COMM2
5	GND	28AWG	9	GND
6	GND	28AWG	8	GND
7	5V0	28AWG	7	VOUT
8	5V0	28AWG	6	VOUT
-	-	28AWG	5*	-
-	-	28AWG	10*	-

(*) There should be cables on ESAD Board J102 mating connector's 5th and 10th pins but these two cables should not be connected to Sensor Board side mating connector. These two cables should be isolated and terminated in the middle. After all the crimps inserted to the housing epoxy should be applied to Harwin / G125-2241096F1 back side.

1.4.4 ESAD Power Interface

There is a power interface between the ESAD and the GE that enables the ESAD to perform its defined functions. The suggested signal characteristics defined for this interface are shown below.

Table 4 ESAD Power Input

Signal Name	Description	Source	Target	Level		
				Min.	Nominal	Max.
GE_POW	Power Input of ESAD	Missile	ESAD	14V	28V	32V

1.4.5 ESAD Discrete Input/Continuity Signal Interface

ESAD has a continuity circuit line and one discrete input lines to control the flow. The table below contains the directions and characteristics of these suggested signals.

Table 5 ESAD Discrete Input and Continuity Signals

Signal Name	Description	Source	Target	Level		
UMB	Continuity	ESAD	ESAD	Resistance	<0.5 Ohm (22-28AWG) 20mA current	
DSC_IN1	Arm Enable PWM Signal 1ms period %50 duty cycle 2 square wave	OC	ESAD	LOW	Min. Voltage	Max. Voltage
					0VDC	0.7VDC
				HIGH	Min. Voltage	Max. Voltage
					2.5V	3.3VDC

1.4.6 Communication Interface

1.4.6.1 RS-485 Datalink Definition

1.4.6.1.1 Physical Layer

ESAD will use RS-485 interface to communicate with the Onboard Computer. Data will be sent and received over RS-485 electrical signals according to the protocol specified in the next section. The specifications of this layer are given in Table 6.

Table 6 Communication Specifications

Spec. Type	Value
Data Bits	8
Parity	None
Start Bit	1
Stop Bit	1
Baudrate	115200bps
Communication Mode	Half Duplex / 100Hz Interval commands

The suggested Bit sequence that is meaningful and descriptive on its own is shown in Table 7. ESAD will send and receive data in the Bit order shown in Table 7. The Bit sequence size is 10 Bits including the start-stop Bits.

Table 7 Bit Sequence

Bit9 (Last Bit)	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0 (First Bit)
Stop Bit	Data Bit(MSB)	Data Bit	Data Bit	Data Bit	Data Bit	Data Bit	Data Bit	Data Bit(LSB)	Start Bit

A differential line will be used for sending and receiving data. If OC is not sending data, it will remain in receive mode.

1.4.6.1.2 RS-485 Bus Message Structure

ESAD will operate in slave mode in accordance with the master/slave concept, waiting for messages to be initiated by the master. ESAD will not send messages unless requested by the OC. Bus management will be done by the OC.

ESAD will be capable of responding to messages sent from the bus in a 100ms period.

The ESAD will have the capability to send the last Bit of the reply message to the bus within 5 ms after the last Bit of the message from the external system is received.

In case the message sent from the external system to the ESAD is interrupted, the ESAD will check the total length of the message to be received, recognize that the message is incomplete and wait 10 ms. If the message is not completed within this time, ESAD does not continue the process and waits for the next command.

The Bit sequence size is 10 Bits including the start-stop Bits. This is the smallest data package. These small packages combine to form larger meaningful packages. The structure of the messages sent by the master or by ESAD will be as shown in Table 8.

Table 8 Message Structure

Message Length	Header	Command	Data	CRC
Min. 5 Byte	1 Byte	1 Byte	0-9 Byte	2 Byte

"Message Length" indicates the length of the sent message in bytes. A message without data should have a value of 5 in the Message Length section. The details of the message will be defined by the finalization of the ESAD electronic board design.

1.4.6.1.3 CRC Calculation

A 16-bit CRC value is added to the end of all ESAD messages . The CRC value is calculated bitwise according to CRC-16 XModem algorithm with initial value 0x0000 and 0x1021 polynomial.The CRC calculation algorithm to be used is as follows.

```
uint16_t calculateCRC(uint8_t* buffer, uint32_t length) {
    uint16_t crc = 0;
    uint16_t data = 0;

    while (0 != length--)
    {
        data = *(buffer++) << 8;
        for (uint32_t i = 0; i < 8; i++)    // Loop for each bit in char
        {
            if ((data ^ crc) & 0x8000)      // Shift CRC, feed back high bit
                crc = (crc << 1) ^ 0x1021;
            else
                crc <<= 1;

            data <<= 1;
        }
    }

    return crc;
}
```

For example, the 16-bit CRC value of the 10-byte sequence consisting of the values [0x10 0x11 0x12 0x13 0x14 0x15 0x16 0x17 0x18 0x19] should be calculated as 0xB987.

1.4.6.2 RS-485 Command & Response (Suggested as Draft)

1.4.6.2.1 Power-On BIT Command (PBIT)

Command Name	PBIT
Command Number	0x11
Description	This is the message sent by OC to request PBIT result information from ESAD. ESAD sends the requested information to the OC with this message.
Pre-Conditions	None
Subsequent Conditions	None

Command Parameters					
Message Length	0x05				
Header	0x01				
Data	Type	Offset	Range	Unit	Description
-	-	-	-	-	-
Notes					
-					
Command Response Parameters					
Message Length	0x07				
Header	0x02				
Data	Type	Offset	Range	Unit	Description
PBITResult	UINT16	0	[0 ; 2^{16}-1]	-	ESAD PBIT Command Response

Bit No*	Description
0	0: PBIT Successful 1: PBIT Unsuccessful
1	0: Discrete Input 1 (arm enable) Not Active 1: Discrete Input 1 (arm enable) Active
2	Reserved
3	0: Safety Switch 1 Not Active 1: Safety Switch 1 Active

						4	0: Dynamic Switch Not Active 1: Dynamic Switch Active
						5	0: Safety Switch 2 Not Active 1: Safety Switch 2 Active
						6	0: Sensor power-up successful 1: Sensor power-up unsuccessful
						7	0: Detonator Connected 1: Detonator Disconnected
						8	0: Umbilical Cord Connected 1: Umbilical Cord Disconnected
						9*	0: Impact Switch Trigger Not Active 1: Impact Switch Trigger Active
						10*	0: Proximity Sensor Trigger Not Active 1: Proximity Sensor Trigger Active
						11	0: Accelerometer Selftest Successful 1: Accelerometer Selftest Unsuccessful
						12	0: High Voltage Not Detected (< 50 V) 1: High Voltage Detected (>= 50 V)
						13	0: CPLD Power up Successful 1: CPLD Power up Unsuccessful
						14-15**	Reserved

* Proximity Sensor and Impact Switch Triggers are Ored on the sensor board, and the FPGA receives them as a single trigger input. 9th Bit 10th Bit will have the same value.

**Bit 0-LSB, Bit 15-MSB

Note
None

1.4.6.2.2 Exchange Command

Command Name	EXCHANGE
Command Number	0x10
Description	It is the command sent by the OC to the ESAD to receive data related to the flight and navigation
Pre-Conditions	None
Subsequent Conditions	None

<table>
<tr><td colspan="6">Command Parameters</td></tr>
<tr><td>Message Length</td><td colspan="5">0x08</td></tr>
<tr><td>Header</td><td colspan="5">0x01</td></tr>
<tr><td>Data</td><td>Type</td><td>Offset</td><td>Range</td><td>Unit</td><td>Description</td></tr>
<tr><td>OC_Status</td><td>UINT8</td><td>0</td><td>[0 ; 2^8-1]</td><td>-</td><td>
<table>
<tr><th>Bit No*</th><th>Description</th></tr>
<tr><td>0</td><td>0: Health Mode
1: Fault Mode</td></tr>
<tr><td>1</td><td>0: Flight Mode
1: Simulation Mode</td></tr>
<tr><td>2</td><td>0: Arm Not Enable
1: Arm Enable</td></tr>
<tr><td>3</td><td>0: Mission Abort Not Enable
1: Mission Abort Enable</td></tr>
<tr><td>4</td><td>0: Arm Abort Not Enable
1: Arm Abort Enable</td></tr>
<tr><td>5</td><td>0: OC Trigger Not Enable
1: OC Trigger Enable</td></tr>
<tr><td>6-7**</td><td>Reserved</td></tr>
</table>
**Bit 0-LSB, Bit 7-MSB
</td></tr>
</table>

SimAccelerationOffset	INT16	1	[0;2^{16}-1]	0.0039g	The most significant 3 bits are reserved. The remaining 13 bits represent a signed two's complement value, with a resolution of 0.0039 g/LSB
Note					
None					
Command Response Parameters					
Message Length	0x0E				
Header	0x02				
Data	Type	Offset	Range	Unit	Description

<table>
<tr>
<td>Flowstatus</td>
<td>UINT16</td>
<td>0</td>
<td>[0 ; 2^{16}-1]</td>
<td>-</td>
<td>
<table>
<tr><th>Bit No*</th><th>Description</th></tr>
<tr><td>0</td><td>0: Arm Counter Not Started
1: Arm Counter Started</td></tr>
<tr><td>1</td><td>0: Flight Counter Not Started
1: Flight Counter Started</td></tr>
<tr><td>2</td><td>0: Flight Mode
1: Simulation Mode</td></tr>
<tr><td>3</td><td>0: Safe Distance Not Exceed
1: Safe Distance Exceed</td></tr>
<tr><td>4</td><td>0: Mission Not Aborted
1: Mission Aborted</td></tr>
<tr><td>5</td><td>0: Detonator Disconnected
1: Detonator Connected</td></tr>
<tr><td>6</td><td>0: Umbilical Cord Disconnected
1: Umbilical Cord Connected</td></tr>
<tr><td>7</td><td>0: Arm Enable Not Received
1: Arm Enable Received</td></tr>
<tr><td>8</td><td>0: Acceleration Not Exceed
1: Acceleration Exceed</td></tr>
<tr><td>9</td><td>0: OC Trigger Not Received
1: OC Trigger Received</td></tr>
<tr><td>10*</td><td>0: Proximity Sensor or Impact Switch Not Triggered
1: Proximity Sensor or Impact Switch Triggered</td></tr>
<tr><td>11</td><td>0: Proximity Sensor Not Powered
1: Proximity Sensor Powered</td></tr>
<tr><td>12</td><td>0: Arm Not Aborted
1: Arm Aborted</td></tr>
<tr><td>13-15**</td><td>Reserved</td></tr>
</table>
* Proximity Sensor and Impact Switch Triggers are Ored on the Proximity Sensor Board, and the FPGA receives them as a single trigger input.
</td>
</tr>
</table>

<table>
<tr><td></td><td></td><td></td><td></td><td></td><td>**Bit 0-LSB, Bit 15-MSB</td></tr>
<tr><td>ArmState</td><td>UINT8</td><td>2</td><td>[0 ; 2^8-1]</td><td>-</td><td>
<table>
<tr><th>Bit No*</th><th>Value & Description</th></tr>
<tr><td>0-2</td><td>0: Safe State
1: Arm Stage1 State
2: Arm Stage2 State
3: Armed State
4: Fired State
5: Fault State</td></tr>
<tr><td>3-7</td><td>Reserved</td></tr>
</table>
*Bit 0-LSB, Bit 7-MSB
</td></tr>
</table>

<table>
<tr><td>FaultStates</td><td>UINT16</td><td>3</td><td>[0 ; 2^{16}-1]</td><td>-</td><td>
<table>
<tr><th>Bit No*</th><th>Description</th></tr>
<tr><td>0</td><td>0: ESAD Health State
1: ESAD Fault State</td></tr>
<tr><td>1</td><td>0: PBIT Successful
1: PBIT Unsuccessful</td></tr>
<tr><td>2</td><td>0: Safe Distance Successful
1: Safe Distance Unsuccessful</td></tr>
<tr><td>3</td><td>0: OC Health State
1: OC Fault State</td></tr>
<tr><td>4</td><td>0: Discrete Arm Enable Not Received
1: Discrete Arm Enable Received**</td></tr>
<tr><td>5</td><td>0: Safety Switch 1 No Error
1: Safety Switch 1 Error</td></tr>
<tr><td>6</td><td>0: Safety Switch 2 No Error
1: Safety Switch 2 Error</td></tr>
<tr><td>7</td><td>0: Dynamic Switch No Error
1: Dynamic Switch Error</td></tr>
<tr><td>8</td><td>0: High Voltage No Error
1: High Voltage Error</td></tr>
<tr><td>9-15*</td><td>Reserved</td></tr>
</table>
*Bit 0-LSB, Bit 15-MSB
** Discrete Arm Received Before Arm Stage 2
</td></tr>
<tr><td>FlightTimer</td><td>UINT16</td><td>5</td><td>[0;2^{16}-1]</td><td>s</td><td>Flight Timer (s)</td></tr>
<tr><td>Acceleration</td><td>INT16</td><td>7</td><td>[0;2^{16}-1]</td><td>0.0039g</td><td>The most significant 3 bits are reserved.
The remaining 13 bits represent a signed two’s complement value, with a resolution of 0.0039 g/LSB</td></tr>
</table>

1.4.6.2.3 Welcome Message

Command Name	WELCOME
Command Number	0xB0
Description	This is the boot message that ESAD sends to the OC at startup.
Pre-Conditions	None
Subsequent Conditions	None

Command Parameters					
Message Length	0x05				
Header	0x01				
Data	Type	Offset	Range	Unit	Description
-	-	-	-	-	-
Note					

Command Response Parameters					
Message Length	0x0C				
Header	0x02				
Data	Type	Offset	Range	Unit	Description
data[1x7]	UINT8	0		-	This is the one-way message information that ESAD sends at startup in ASCII format (“WELCOME”)
Note					
None					

1.4.6.2.4 Version Command

Command Name	VERSION
Command Number	0x13
Description	This is the message sent by the OC to request embedded firmware version information from ESAD.

Pre-Conditions	None
Subsequent Conditions	None

Command Parameters					
Message Length	0x05				
Header	0x01				
Data	Type	Offset	Range	Unit	Description
-	-	-	-	-	-
Note					
None					

Command Response Parameters					
Message Length	0x0B				
Header	0x02				
Data	Type	Offset	Range	Unit	Description
version[1x6]	UINT8	0	[0,255]	-	Format; **XXXYYY** XXX: FPGA Firmware version (HEX) YYY: CPLD Firmware version (HEX) It will be terminated as \0
Note					
None					

1.4.6.2.5 HighVoltage Command

Command Name	HIGHVOLTAGE
Command Number	0x12
Description	This is the message sent by the OC to request high voltage value and states information from ESAD.
Pre-Conditions	None
Subsequent Conditions	None

Command Parameters					
Message Length	0x05				
Header	0x01				
Data	Type	Offset	Range	Unit	Description
-	-	-	-	-	-
Note					
None					

Command Response Parameters					
Message Length	0x08				
Header	0x02				
Data	Type	Offset	Range	Unit	Description
High_Voltage_Flags	UINT8	0	[0 ; 2^8-1]	-	

Bit No*	Description
0	0: SS1 Not Enabled 1: SS1 Enabled
1	0: SS1 Not Sensed 1: SS1 Sensed
2	0: SS2 Not Enabled 1: SS2 Enabled
3	0: SS2 Not Sensed 1: SS2 Sensed
4	0: DS Not Enabled

<table>
<tr><td></td><td></td><td></td><td></td><td></td><td>
<table>
<tr><td></td><td>1: DS Enabled</td></tr>
<tr><td>5</td><td>0: DS Not Sensed
1: DS Sensed</td></tr>
<tr><td>6</td><td>0: Charger Not Enabled
1: Charger Enabled</td></tr>
<tr><td>7</td><td>0: HV Not Charged
1: HV Charged</td></tr>
</table>
*Bit 0-LSB, Bit 15-MSB
</td></tr>
<tr><td>High_Voltage_Value</td><td>UINT16</td><td>1</td><td>[0,1800]</td><td>V</td><td>Resolution : 1V/LSB</td></tr>
<tr><td colspan="6">Note</td></tr>
<tr><td colspan="6">None</td></tr>
</table>

1.4.7 Mechanical Interface

Since the payload and the ESAD will be located inside a sealed enclosure on the main drone platform, it has been decided that the ESAD board will be mounted directly inside that closed payload compartment on the drone, without an additional mechanical housing. The ESAD board dimensions are shown below, and the screw/fastening standard to be used for the mechanical interface will be agreed.

1.4.7.1 ESAD Board Mounting Interfaces

There are three M3 mounting holes on ESAD Board for the mechanical connection interface. The other 3 big holes on ESAD board is reserved for ARTRON and should not be used by USER.

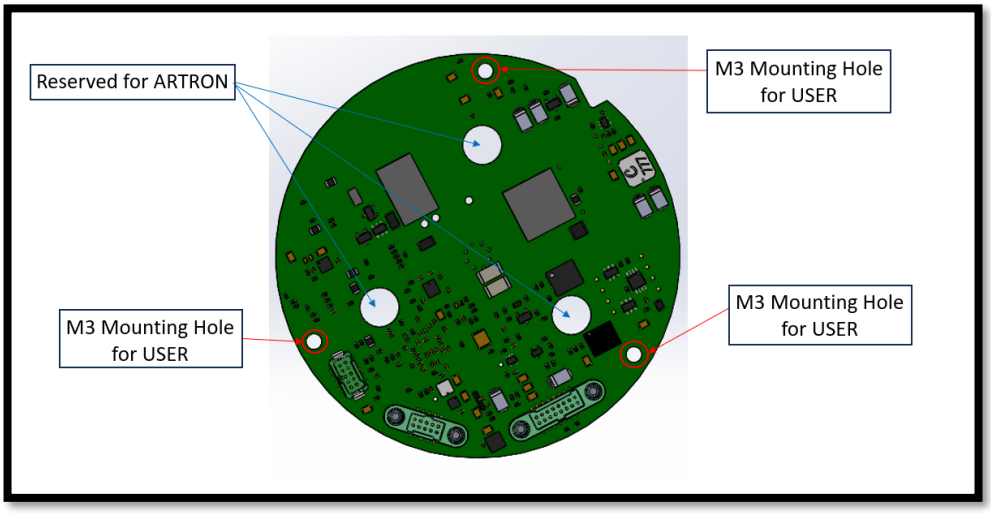


Figure 10 Mounting Interfaces on ESAD Board

1.4.7.2 Sensor Board Mounting Interfaces

There are two M2 mounting holes on Sensor Board for the mechanical connection interface.

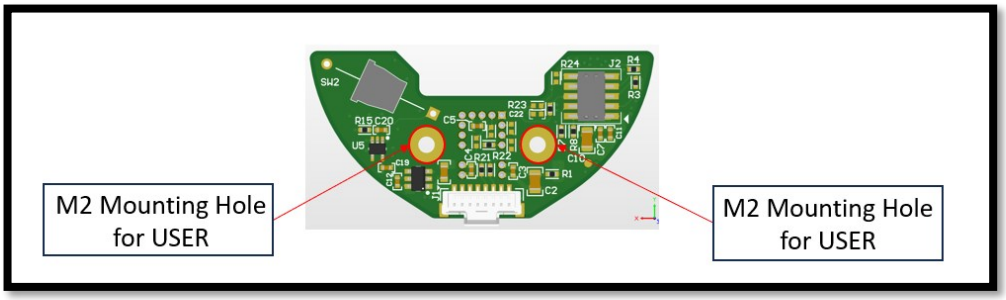


Figure 11 Mounting Interfaces on Sensor Board

The following figures shows the detonator integration.

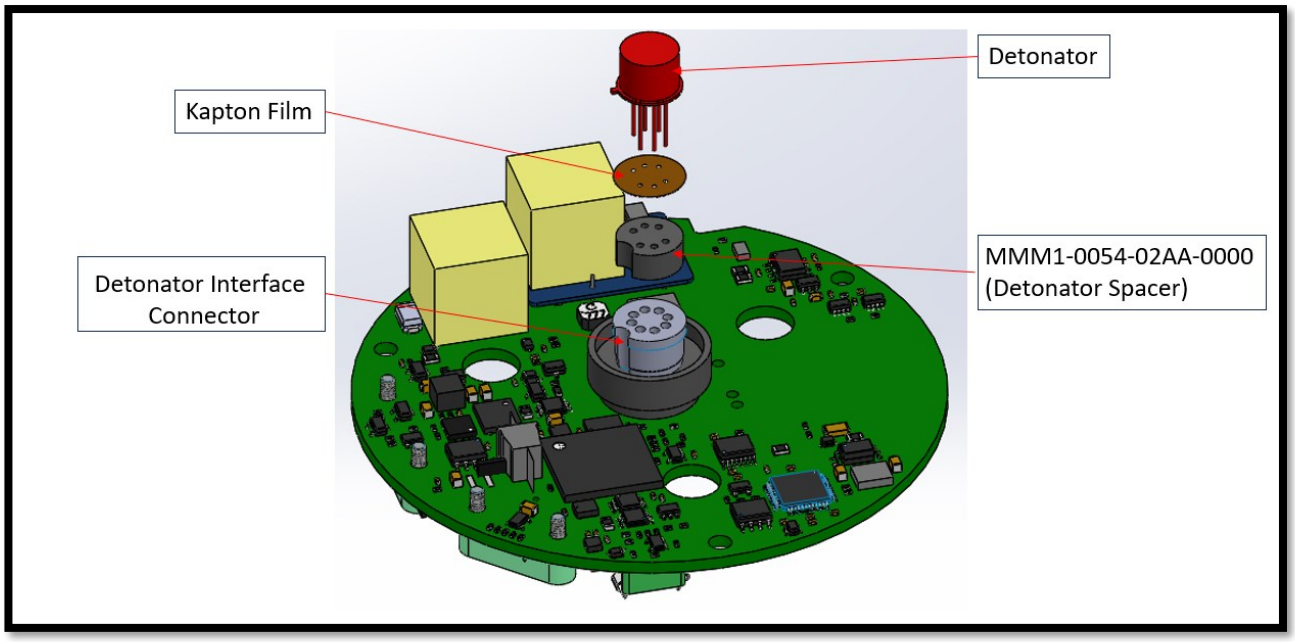


Figure 12: Detonator Integration

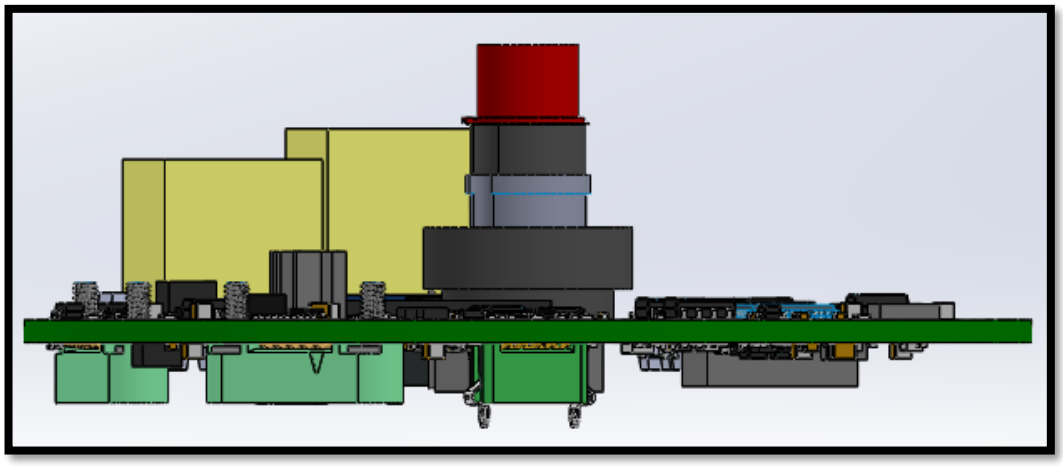


Figure 13: Detonator Integration

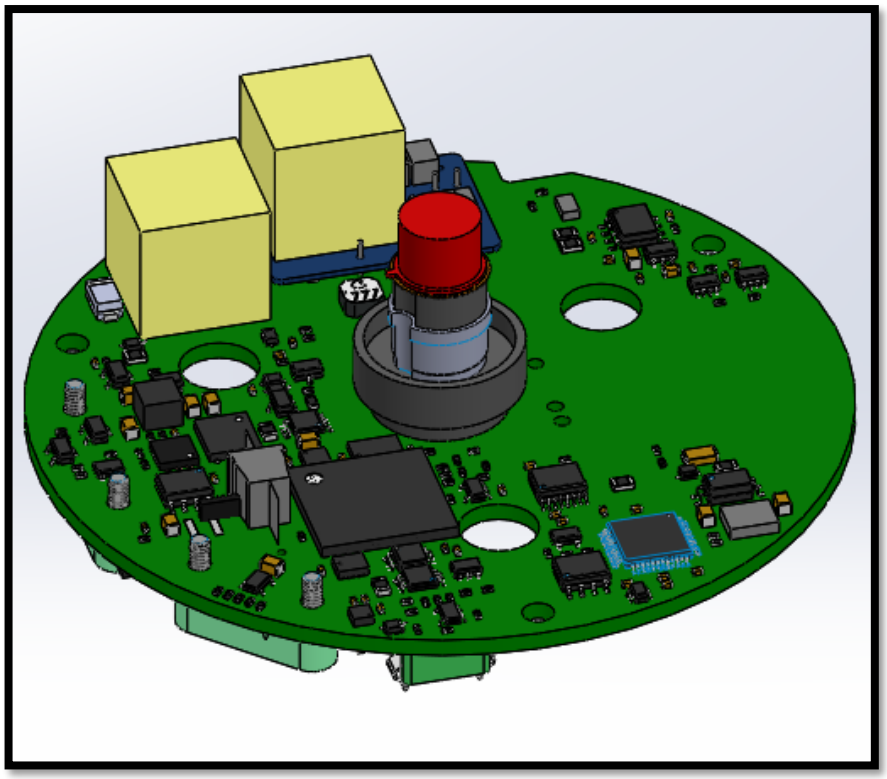


Figure 14: Detonator Integration

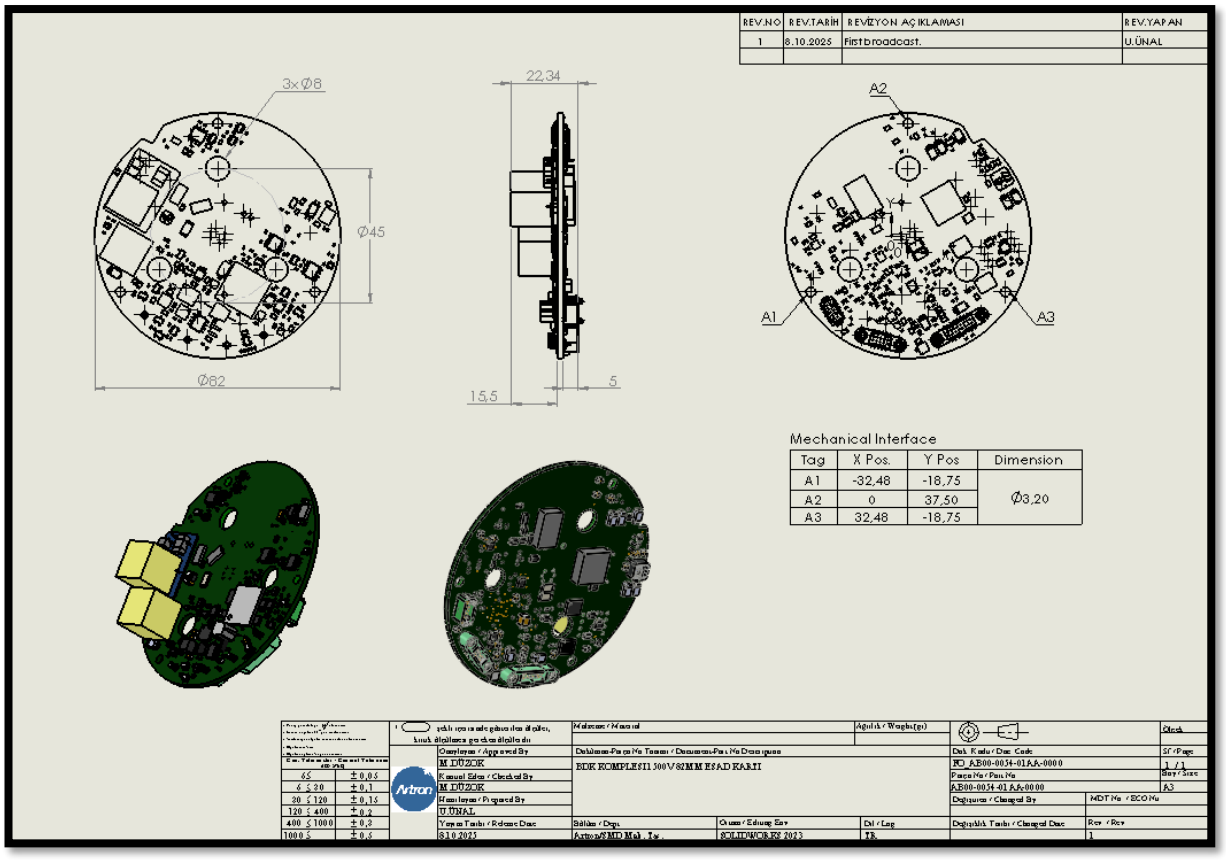


Figure 15: ESAD Board Pyhsical Characteristics

1.4.7.3 Proximity Sensor Board Dimension

In addition to ESAD unit, the proximity sensor board is given in below in**Error! Reference source n ot found.**. This board should be placed onto the front face of the upper system.

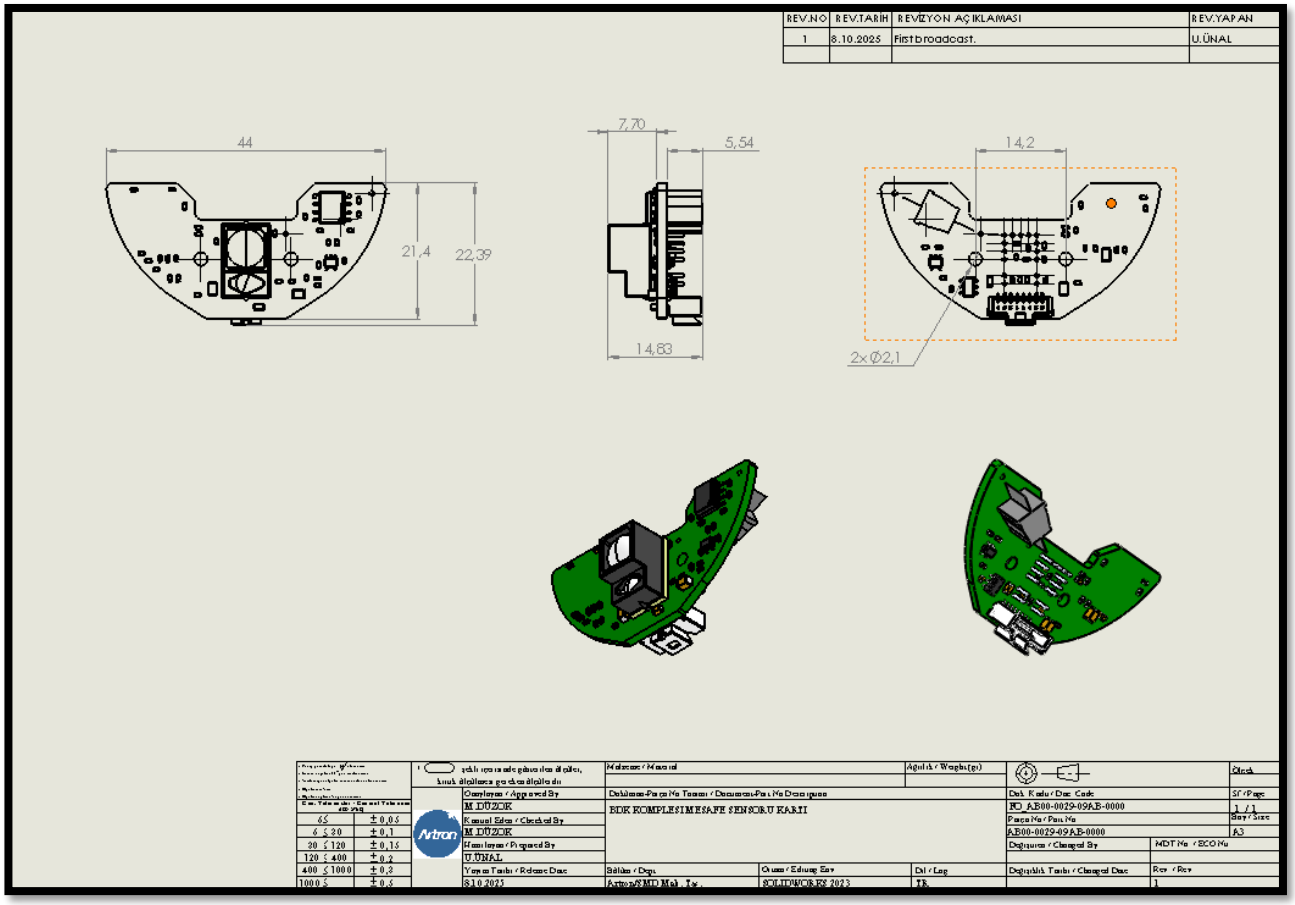
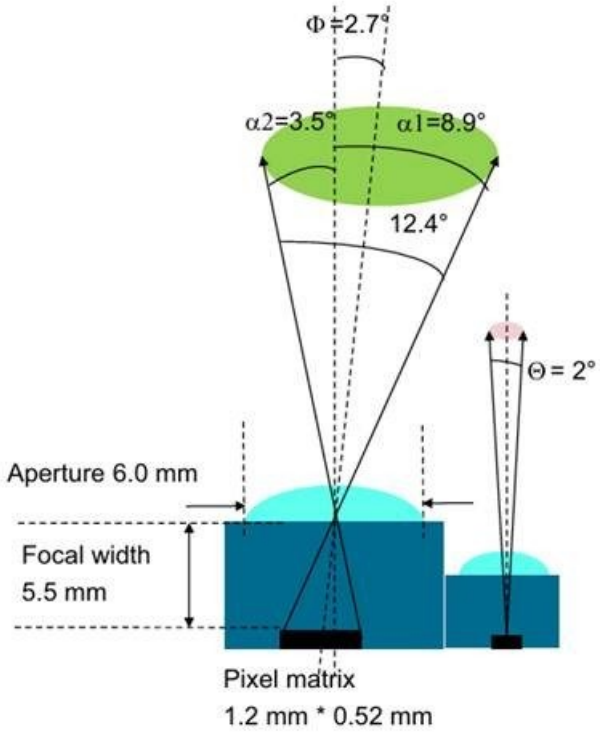


Figure 16 Proximity Sensor Board Pyhsical Characteristics

The field of view (FOV) that must be maintained from the sensor lens vertically and horizontally is as follows.



Rx cone (typ.):
12.4° x 5.4° divergence
2.7° squint angle

Tx cone (typ.):
2° x 2° divergence
No squint angle